K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

DATABASE MANAGEMENT SYSTEMS
[ENCS204]

Bachelor of Technology
in
Computer Science and Engineering

Submitted by
[LAKSHITA KALRA](2401010117)

Course Teacher:
MISS MANSI KAJAL

School of Engineering and Technology

K.R.MANGALAM UNIVERSITY
Sohna,Haryana 122103,India
JAN- MAY[2026]

LAB ASSIGNMENT- 4

AIM:

To perform user creation, deletion, and privilege management using SQL commands in MySQL.

DESCRIPTION:

In this experiment, we learn how to:

- Create new users in database
- Give permissions (GRANT)
- Remove permissions (REVOKE)
- Delete users
- Check users and their privileges

This is important for **database security** so that only authorized users can access data.

PROCEDURE:

Step 1: Open MySQL Workbench

1. Open MySQL Workbench
2. Click on your local connection (localhost)
3. Enter your root password
4. Click OK → SQL Editor will open
5. Create Database college
6. Use college

Step 2: Create Users

```
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123';
CREATE USER 'user2'@'localhost' IDENTIFIED BY 'password456';
```

1	11:54:15	CREATE USER 'user1'@'localhost' IDENTIFIED B...	0 row(s) affected	0.020 sec
2	11:54:15	CREATE USER 'user2'@'localhost' IDENTIFIED...	0 row(s) affected	0.0034 sec

Step 3: Grant Privileges

GRANT SELECT, INSERT ON Student TO 'user1'@'localhost';
GRANT ALL PRIVILEGES ON Course TO 'user2'@'localhost';
GRANT SELECT ON Student TO 'user1'@'localhost' WITH GRANT OPTION;

9	13:18:59	GRANT SELECT, INSERT ON Student TO 'user1'@'localhost'	0 row(s) affected	0.0070 sec
10	13:18:59	GRANT ALL PRIVILEGES ON Course TO 'user2'@'localhost'	0 row(s) affected	0.0011 sec
11	13:18:59	GRANT SELECT ON Student TO 'user1'@'localhost' WITH GRANT OPTION	0 row(s) affected	0.00085 sec

Step 4: Revoke Privileges

REVOKE INSERT ON Student FROM 'user1'@'localhost';
REVOKE ALL PRIVILEGES ON Course FROM 'user2'@'localhost';

12	13:21:37	REVOKE INSERT ON Student FROM 'user1'@'localhost'	0 row(s) affected	0.0030 sec
13	13:21:37	REVOKE ALL PRIVILEGES ON Course FROM 'user2'@'localhost'	0 row(s) affected	0.00091 sec

Step 5: Delete User

DROP USER 'user2'@'localhost';

14	13:22:44	DROP USER 'user2'@'localhost'	0 row(s) affected	0.0069 sec

Step 6: Check Users

SELECT User, Host FROM mysql.user;

100% 35:1

Result Grid Filter Rows: Search Edit: Export/Import:

User	Host
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost

user 1 Apply Revert

Action Output

	Time	Action	Response	Duration / Fetch Time
6	13:17:55	CREATE TABLE Student (id INT PRIMARY KEY, name VARCHAR(50), marks I...	0 row(s) affected	0.035 sec
7	13:18:03	CREATE TABLE Course (course_id INT PRIMARY KEY, course_name VARCHAR(...	0 row(s) affected	0.010 sec
8	13:18:11	CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123'	Error Code: 1396. Operation CREATE USER failed for '...	0.015 sec
9	13:18:59	GRANT SELECT, INSERT ON Student TO 'user1'@'localhost'	0 row(s) affected	0.0070 sec
10	13:18:59	GRANT ALL PRIVILEGES ON Course TO 'user2'@'localhost'	0 row(s) affected	0.0011 sec
11	13:18:59	GRANT SELECT ON Student TO 'user1'@'localhost' WITH GRANT OPTION	0 row(s) affected	0.00085 sec
12	13:21:37	REVOKE INSERT ON Student FROM 'user1'@'localhost'	0 row(s) affected	0.0030 sec
13	13:21:37	REVOKE ALL PRIVILEGES ON Course FROM 'user2'@'localhost'	0 row(s) affected	0.00091 sec
14	13:22:44	DROP USER 'user2'@'localhost'	0 row(s) affected	0.0069 sec
15	13:23:33	SELECT User, Host FROM mysql.user LIMIT 0, 1000	5 row(s) returned	0.0056 sec / 0.00001...

Step 7: Check Privileges

SHOW GRANTS FOR 'user1'@'localhost';

100% 37:1

Result Grid Filter Rows: Search Export:

Grants for user1@localhost
GRANT USAGE ON *.* TO `user1`@`localhost`
GRANT SELECT ON `college`.`student` TO `us...

Result 2 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
7	13:18:03	CREATE TABLE Course (course_id INT PRIMARY KEY, course_name VARCHAR(...	0 row(s) affected	0.010 sec
8	13:18:11	CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123'	Error Code: 1396. Operation CREATE USER failed for '...	0.015 sec
9	13:18:59	GRANT SELECT, INSERT ON Student TO 'user1'@'localhost'	0 row(s) affected	0.0070 sec
10	13:18:59	GRANT ALL PRIVILEGES ON Course TO 'user2'@'localhost'	0 row(s) affected	0.0011 sec
11	13:18:59	GRANT SELECT ON Student TO 'user1'@'localhost' WITH GRANT OPTION	0 row(s) affected	0.00085 sec
12	13:21:37	REVOKE INSERT ON Student FROM 'user1'@'localhost'	0 row(s) affected	0.0030 sec
13	13:21:37	REVOKE ALL PRIVILEGES ON Course FROM 'user2'@'localhost'	0 row(s) affected	0.00091 sec
14	13:22:44	DROP USER 'user2'@'localhost'	0 row(s) affected	0.0069 sec
15	13:23:33	SELECT User, Host FROM mysql.user LIMIT 0, 1000	5 row(s) returned	0.0056 sec / 0.00001...
16	13:25:36	SHOW GRANTS FOR 'user1'@'localhost'	2 row(s) returned	0.0025 sec / 0.00000...

STEP 8: Show All Users

SELECT User, Host FROM mysql.user;

100% 35:1

Result Grid Filter Rows: Search Edit: Export/Import:

User	Host
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost

user 3 Apply Revert

Action Output

	Time	Action	Response	Duration / Fetch Time
8	13:18:11	CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password123'	Error Code: 1396. Operation CREATE USER failed for '...	0.015 sec
9	13:18:59	GRANT SELECT, INSERT ON Student TO 'user1'@'localhost'	0 row(s) affected	0.0070 sec
10	13:18:59	GRANT ALL PRIVILEGES ON Course TO 'user2'@'localhost'	0 row(s) affected	0.0011 sec
11	13:18:59	GRANT SELECT ON Student TO 'user1'@'localhost' WITH GRANT OPTION	0 row(s) affected	0.00085 sec
12	13:21:37	REVOKE INSERT ON Student FROM 'user1'@'localhost'	0 row(s) affected	0.0030 sec
13	13:21:37	REVOKE ALL PRIVILEGES ON Course FROM 'user2'@'localhost'	0 row(s) affected	0.00091 sec
14	13:22:44	DROP USER 'user2'@'localhost'	0 row(s) affected	0.0069 sec
15	13:23:33	SELECT User, Host FROM mysql.user LIMIT 0, 1000	5 row(s) returned	0.0056 sec / 0.00001...
16	13:25:36	SHOW GRANTS FOR 'user1'@'localhost'	2 row(s) returned	0.0025 sec / 0.00000...
17	13:26:45	SELECT User, Host FROM mysql.user LIMIT 0, 1000	5 row(s) returned	0.0012 sec / 0.00000...

CONCLUSION:

In this experiment, I successfully learned and implemented SQL user management and access control in a database system. I created users, assigned privileges using GRANT commands, modified permissions using REVOKE commands, and deleted users using DROP USER. I also verified user details and their privileges using system queries.

This helped us understand how database security is maintained by controlling user access, ensuring that only authorized users can perform specific operations on tables.